

HOW LIE DETECTORS WORK?

Criminals telling lies become a serious problem for the law enforcement authorities. In that case, they are put to lie detector tests. Naturally, curiosity arises to know about the working of lie detectors

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Lie detectors do not, as their nickname suggests, detect lies. Polygraphs, commonly called 'lie detectors,' are electronic instruments that monitor a person's physiological reactions to a variety of questions by the examiners. These can only detect whether the deceptive behavior is being displayed.

A polygraph instrument is basically a combination of medical devices that monitor changes occurring in the body by using electronic sensors. As a person is questioned about a certain event or incident, the examiner sees how the person's heart rate, blood pressure, respiratory rate and electro-dermal activity (sweat in fingers) change in comparison to normal levels. Fluctuations may indicate that the person is being deceptive, but exam results are open to interpretation by the examiner.

Polygraph: the method of testing

Polygraph exams are most often associated with criminal investigations, but there are other instances too in which these are used. In the future, one may be subjected to a polygraph exam before being hired for a job. Polygraph examinations are designed to look for significant involuntary responses going on in a person's body when he is subjected to stress, such as the stress associated with deception.

The exams are unable to specifically detect whether a person is lying, according to polygrapher Dr Bob Lee,

former executive director of operations at Axciton Systems (USA), a manufacturer of polygraph instruments. But there are certain physiological re-

sponses that most of us undergo when attempting to deceive another person. By questioning a subject about a particular issue under investigation and

examining his physiological reactions to those questions, a polygraph examiner can determine deceptive behaviour.

What does a lie detector consist of?

Fig. 1 shows the circuit of a lie detector. It consists of three transistors (T1 to T3), a capacitor (C1), two lights or LEDs (L1 and L2), five

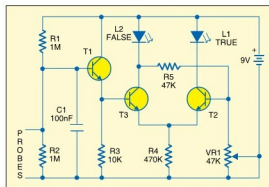


Fig. 1: Circuit of the lie detector

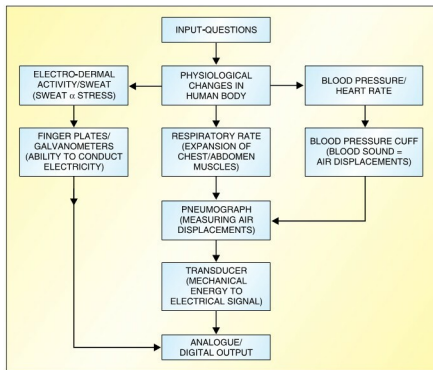


Fig. 2: Block diagram of lie detector indicating different sections of lie detection and the deceptive behaviour